

C program to create arrays of sizes 10^3 , 10^4 , and 10^5 , compute the sum of their elements, and measure the time taken to perform the summation.

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <time.h>
```

```
void test_array(int n) {
```

```
    int *arr = (int *)malloc(n * sizeof(int));
```

```
    long long sum = 0;
```

```
    for(int i = 0; i < n; i++) {
```

```
        arr[i] = i + 1;
```

```
    }
```

```
    clock_t start = clock();
```

```
    for(int i = 0; i < n; i++) {
```

```
        sum += arr[i];
```

```
    }
```

```
    clock_t end = clock();
```

```
    double time_taken = (double)(end - start) / CLOCKS_PER_SEC;
```

```
    printf("Array Size: %d\n", n);
```

```
    printf("Sum = %lld\n", sum);
```

```
    printf("Time taken = %f seconds\n\n", time_taken);
```

```
    free(arr);  
}  
  
int main() {  
    test_array(1000);  
    test_array(10000);  
    test_array(100000);  
    return 0;  
}
```

OUTPUT:

```
arpita@DESKTOP-9FJDKRQ:~$ nano test.c  
arpita@DESKTOP-9FJDKRQ:~$ gcc -O2 test.c -o test  
arpita@DESKTOP-9FJDKRQ:~$ ./test  
Array Size: 1000  
Sum = 500500  
Time taken = 0.000003 seconds  
  
Array Size: 10000  
Sum = 50005000  
Time taken = 0.000011 seconds  
  
Array Size: 100000  
Sum = 5000050000  
Time taken = 0.000095 seconds
```

C program to compare the execution time of row-wise and column-wise traversal of a 2D array for different matrix sizes.

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <time.h>
```

```
void test(int n) {
```

```
    int (*arr)[n] = malloc(sizeof(int[n][n]));
```

```
    for(int i = 0; i < n; i++)
```

```
        for(int j = 0; j < n; j++)
```

```
            arr[i][j] = i + j;
```

```
    volatile int temp;
```

```
    clock_t start_row = clock();
```

```
    for(int i = 0; i < n; i++)
```

```
        for(int j = 0; j < n; j++)
```

```
            temp = arr[i][j];
```

```
    clock_t end_row = clock();
```

```
    clock_t start_col = clock();
```

```
    for(int j = 0; j < n; j++)
```

```
        for(int i = 0; i < n; i++)
```

```
            temp = arr[i][j];
```

```
    clock_t end_col = clock();
```

```

double row_time = (double)(end_row - start_row) / CLOCKS_PER_SEC;
double col_time = (double)(end_col - start_col) / CLOCKS_PER_SEC;

printf("Matrix Size: %d x %d\n", n, n);
printf("Row-wise access time : %f seconds\n", row_time);
printf("Column-wise access time: %f seconds\n", col_time);
printf("Difference      : %f seconds\n\n", col_time - row_time);

free(arr);
}

int main() {
    test(100);
    test(1000);
    return 0;
}

```

OUTPUT:

```

arpita@DESKTOP-9FJDKRQ:~$ nano twod.c
arpita@DESKTOP-9FJDKRQ:~$ gcc -O2 twod.c -o twod
arpita@DESKTOP-9FJDKRQ:~$ ./twod
Matrix Size: 100 x 100
Row-wise access time   : 0.000011 seconds
Column-wise access time: 0.000015 seconds
Difference              : 0.000004 seconds

Matrix Size: 1000 x 1000
Row-wise access time   : 0.000944 seconds
Column-wise access time: 0.001863 seconds
Difference              : 0.000919 seconds

```